CLINICAL · NECK, THYROID & SALIVARY

Paragangliomas, neural tumours and vascular anomalies

A slow-growing, painless lump high in the neck that moves from side to side but not up and down, sometimes with a bruit, is a carotid body tumour until the scan says otherwise. Never needle it. Paragangliomas are often hereditary: every patient is tested for SDH mutations and screened for catecholamine secretion. In children, the vascular anomalies are sorted by one question. A haemangioma is a tumour that grows after birth and then involutes. A malformation is present at birth and grows with the child.

AT A GLANCE

- Carotid body tumour splays the internal and external carotids (lyre sign).
- Up to 40% of paragangliomas are hereditary (SDH).
- Vagal tumours push the carotids forward without splaying.
- Haemangiomas involute; malformations grow with the child.

1 CHAPTER ONE Paragangliomas

Tumours of the paraganglia.

1.1 Biology

Paragangliomas arise from the neural-crest *paraganglia*, the chemoreceptor tissue of the carotid body, the vagal ganglia, the jugular bulb and the middle ear. Histologically they are nests of *chief (type I) cells* surrounded by *sustentacular (type II) cells*, the pattern called *Zellballen* ("balls of cells"), in a highly vascular stroma. The carotid body is the commonest site in the head and neck, accounting for about half to 60%, followed by the jugular foramen (glomus jugulare), the middle ear (glomus tympanicum) and the vagus (glomus vagale, about 5 to 10%).

Up to 40% are hereditary, mostly from mutations in the *succinate dehydrogenase* genes SDHB, SDHC, SDHD and others. Hereditary tumours appear at a younger age, are more often multiple and bilateral (about a third), and come with

phaeochromocytomas; SDHB carriers have the highest risk of malignancy. *Genetic testing is therefore offered to every patient.* Chronic high-altitude hypoxia, by contrast, causes carotid body hyperplasia.

About 5 to 10% of carotid tumours, and more vagal tumours, are malignant, and malignancy is defined only by metastasis to nodes or distant sites, not by histology or local invasion. Only 1 to 3% secrete catecholamines, but everyone is screened with *plasma free metanephrines* or 24-hour urinary fractionated metanephrines, both to find a functioning tumour and to find a phaeochromocytoma before anaesthesia.

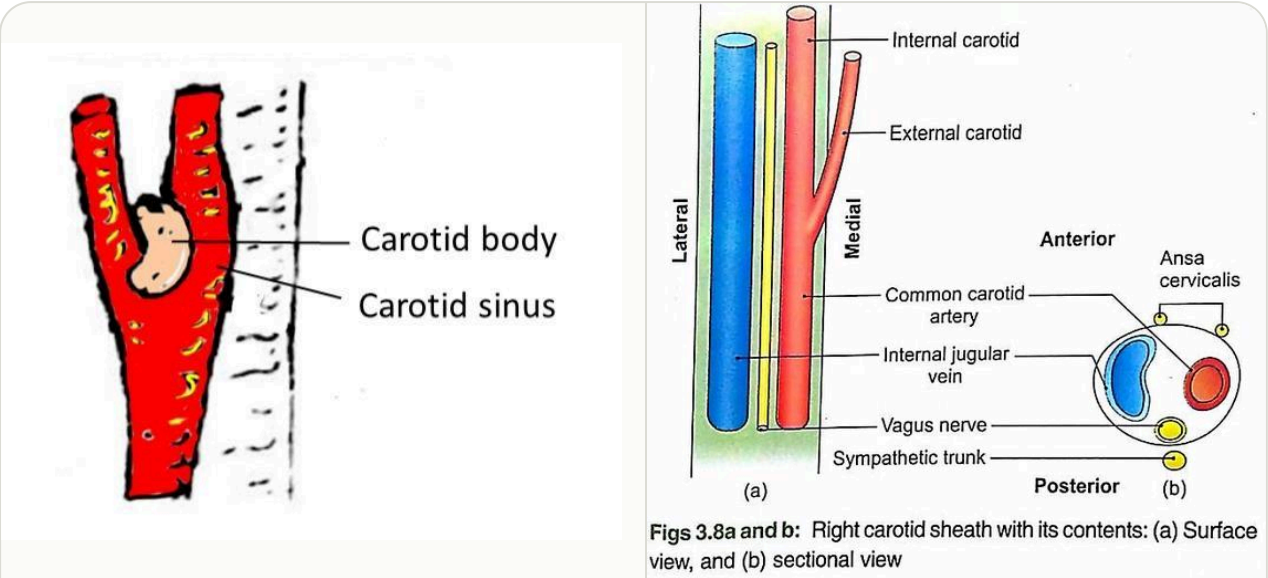


FIGURE 1 The carotid body at the bifurcation, and the carotid sheath and its contents.
Anatomy atlas · Doss p. 137, ENT Steps p. 266

1.2 Carotid body tumour

The carotid body tumour is a painless, slowly enlarging mass at the *carotid bifurcation*, at the anterior border of the sternomastoid at the level of the hyoid. It grows about 1 mm a year, doubling in four to five years. It moves from side to side but *not vertically* (*Fontaine's sign*), transmits or has its own pulsation, may have a bruit, and may bulge into the oropharynx. Cranial nerve palsies of X, XII or the sympathetic chain are uncommon before surgery, and pain or rapid growth suggests malignancy.

Contrast CT or MRI shows an intensely enhancing mass *splaying the internal and external carotids* apart. On MRI it has a "*salt-and-pepper*" pattern, haemorrhage providing the salt and flow voids the pepper, and angiography shows the *lyre sign* of splayed carotids around a vascular blush. Functional imaging with Ga-68 DOTATATE

PET looks for multiple tumours, and if carotid sacrifice may be needed, the circle of Willis and collateral flow are assessed by balloon occlusion testing.

NO NEEDLE, NO BIOPSY

FNA of a suspected carotid body tumour is avoided, and open or core biopsy condemned. They bleed, they rarely give a diagnosis, and imaging is characteristic.

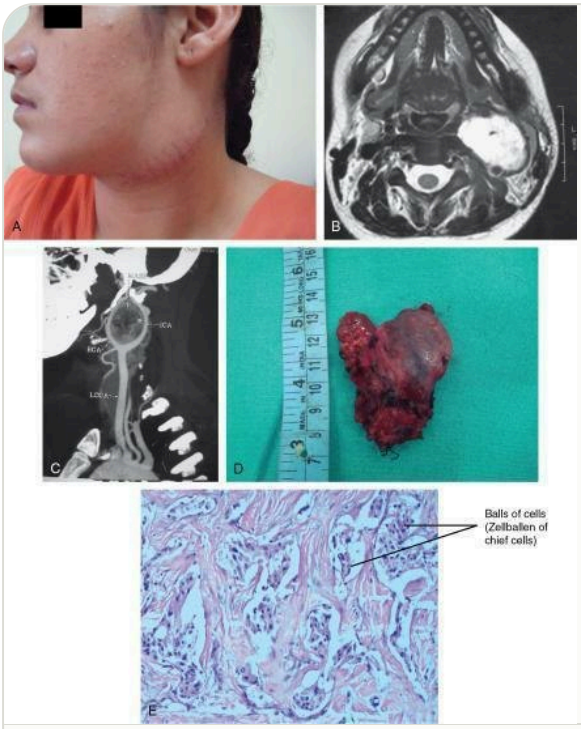


FIGURE 2 Carotid body tumour: (A) clinical swelling; (B) MRI; (C, D) angiography showing splaying of the internal and external carotids (lyre sign) with tumour blush; (E) the excised specimen. Dhingra, fig. 77.6

SHAMBLIN CLASSIFICATION

GROUP	RELATION TO THE CAROTIDS	SURGICAL MEANING
I	Small, localised, easily separated from the vessels	Low risk
II	Adherent to or partly surrounding the vessels	Careful subadventitial dissection; nerve injury more likely
III	Completely encasing the carotids	May need carotid resection and grafting; highest risk of stroke and nerve injury

Surgery is the only cure and is preferred for younger, fit patients and for tumours that are growing, secreting or causing symptoms. Older or frail patients with small,

stable tumours can be observed with serial MRI. Conventional or stereotactic radiotherapy halts growth in about 90% of unresectable tumours, unfit patients and some bilateral cases, though the tumour does not disappear.

1. Transcervical approach, beginning with a level II to III dissection.
2. Control of the common, internal and external carotids.
3. Identification of the vagus, hypoglossal and superior laryngeal nerves and the sympathetic chain.
4. Dissection in the periadventitial plane (the "white line"), working from below upwards.
5. The external carotid may be ligated.
6. A vascular surgeon should be available for reconstruction.

Some surgeons embolise large tumours beforehand, accepting a risk of stroke and nerve palsy. After surgery, cranial nerve deficits of the vagus, hypoglossal and sympathetic chain are common with large tumours, though many recover, and stroke is the feared complication. *First bite syndrome*, a sharp parotid pain with the first bites of a meal, follows loss of the sympathetic supply to the parotid. Resection of both carotid bodies causes *baroreflex failure* with labile blood pressure and tachycardia, so bilateral tumours are staged, or one is observed.

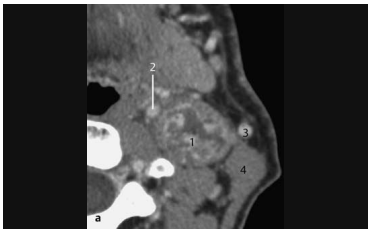
OPERATIVE GUIDES IN THEATRE 1 operation

Excision of carotid body tumours.

Excision of Carotid Body Tumors
Carotid body tumour

IMAGING IN THE APP 1 case

Carotid body paraganglioma.



Carotid Body Paraganglioma
Tumours · 13 images

1.3 Vagal paraganglioma

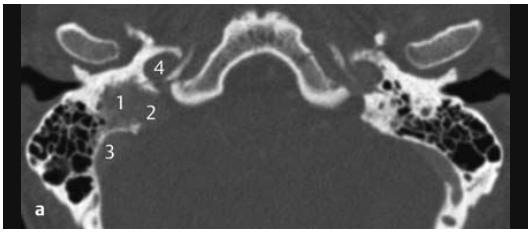
The vagal paraganglioma usually arises from the inferior (nodose) ganglion, high in the parapharyngeal space. It displaces both carotids *anteromedially* without splaying them, separates the internal jugular vein from the carotid, and may extend to the jugular foramen and intracranially. About a third of patients already have vagal weakness and hoarseness, and malignancy is commoner than with carotid tumours. Because surgery almost always sacrifices the vagus, small tumours with normal nerve function are observed or irradiated, and surgery is kept for growing tumours and for those that have already paralysed the nerve.

1.4 Jugular and tympanic paragangliomas

Glomus jugulare and tympanicum tumours present with *pulsatile tinnitus*, conductive hearing loss and a red mass behind the drum, the *rising sun* sign, which blanches on pneumatic pressure (*Brown's sign*), and jugulare tumours may cause lower cranial nerve palsies, the *jugular foramen syndrome*. They are classified by Fisch or by Glasscock–Jackson and treated by surgery through transcanal, mastoid or infratemporal approaches, by radiotherapy, or by observation.

IMAGING IN THE APP 1 case

Glomus jugulare paraganglioma.



Glomus Jugulare Tumor (Paraganglioma)

Tumours · 11 images

OPERATIVE GUIDES IN THEATRE 1 operation

Glomus tumours.

Glomus Tumors

Glomus tumours

2 CHAPTER TWO Neural tumours

Schwannomas and neurofibromas.

2.1 Schwannoma (neurilemmoma)

The schwannoma is a benign, encapsulated tumour of Schwann cells that grows *eccentrically* on its nerve, so the fascicles can often be separated from it. Histology shows cellular *Antoni A* areas with palisading *Verocay bodies* and loose *Antoni B* areas.

Most head and neck schwannomas arise in the *parapharyngeal space*, where they are the second commonest tumour after salivary tumours, usually from the vagus or the cervical sympathetic chain, but also from IX, XI, XII and the cervical and brachial plexuses. They present as a slow, painless lateral neck or parapharyngeal mass that moves from side to side but not vertically, and pressure may send pain along the nerve. The nerve of origin usually still works.

On imaging they enhance less intensely than a paraganglioma, have *no flow voids*, and show a target sign on T2. A *vagal* schwannoma pushes the carotids forward and separates the carotid from the jugular vein, while a *sympathetic* schwannoma pushes both carotids forward without separating the vessels. They are removed by intracapsular enucleation preserving the fascicles, but deficits remain common afterwards: hoarseness after vagal tumours, and *Horner syndrome* and first bite syndrome after sympathetic tumours. Small tumours in older patients can be observed.

2.2 Neurofibroma

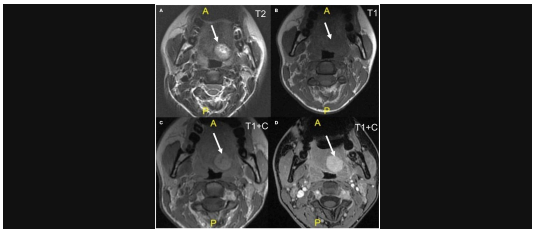
The neurofibroma is unencapsulated and intertwined with its nerve, so the nerve has to be sacrificed to remove it. Multiple and plexiform neurofibromas are the hallmark of neurofibromatosis type 1, and rapid growth or pain suggests malignant change to a *malignant peripheral nerve sheath tumour*.

DISTINGUISHING PARAPHARYNGEAL AND CAROTID SPACE MASSES

FEATURE	CAROTID BODY TUMOUR	VAGAL PARAGANGLIOMA	VAGAL SCHWANNOMA	SYMPATHETIC SCHWANNOMA	PLEOMORPHIC ADENOMA (PRESTYLOID)
Position of the carotids	Splayed apart at the bifurcation	Pushed anteromedially, not splayed	Pushed anteriorly	Pushed anteriorly, vein not separated	Pushed posteriorly
Flow voids	Yes	Yes	No	No	No
Nerve deficit before surgery	Rare	Vagal in about a third	Uncommon	Horner occasionally	None

IMAGING IN THE APP 1 case

Schwannomas of the oral cavity and neck.



Schwannomas of the oral cavity and neck

After surgery · 3 images

OPERATIVE GUIDES IN THEATRE 2 operations

Excision of neural tumours, and tumours of the prestyloid parapharyngeal space.

Excision of Neural Tumors of the Parapharyngeal Space

Neural tumours

Management of Tumors in the Prestyloid Parapharyngeal Space

Parapharyngeal space

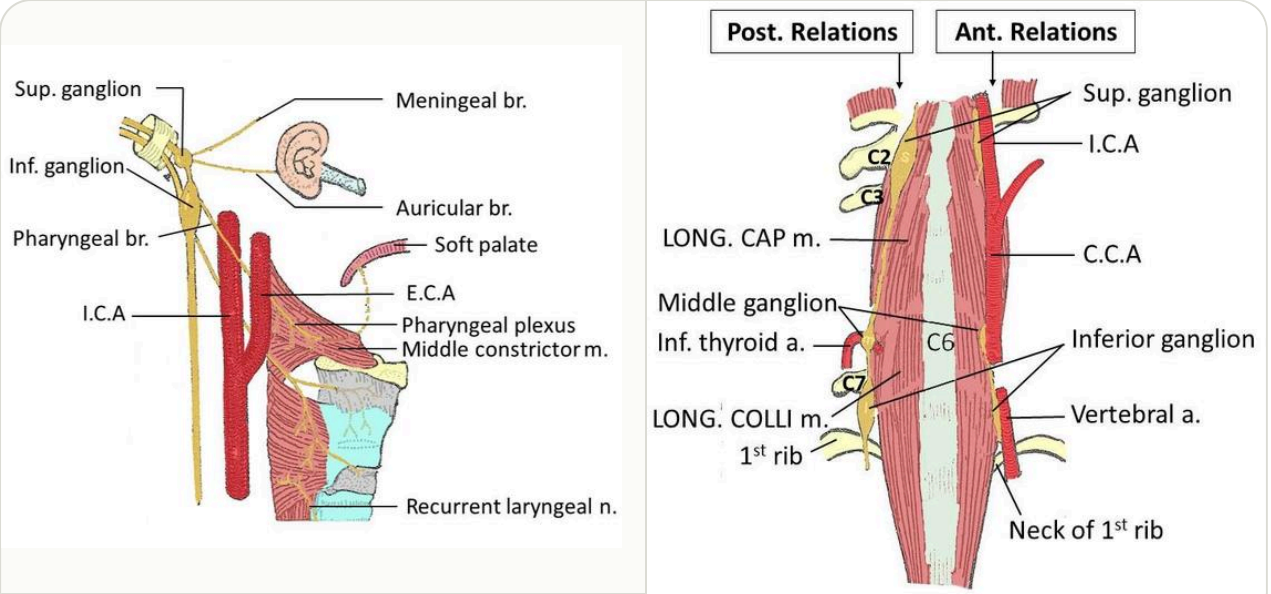


FIGURE 3 The vagus nerve and the cervical sympathetic chain in the neck.
Anatomy atlas · Doss p. 153, Doss p. 160

CHAPTER THREE

3 Vascular anomalies

Tumours and malformations.

3.1 Classification (ISSVA)

HAEMANGIOMA VERSUS VASCULAR MALFORMATION

FEATURE	INFANTILE HAEMANGIOMA	VASCULAR MALFORMATION
Nature	True tumour (endothelial proliferation)	Error of vessel development
At birth	Absent or a faint precursor mark	Present (may not be visible)
Course	Proliferates for about a year, then involutes over years	Grows in proportion with the child; never involutes; growth spurts at puberty, pregnancy, infection, trauma
Marker	GLUT-1 positive	GLUT-1 negative

FEATURE	INFANTILE HAEMANGIOMA	VASCULAR MALFORMATION
Treatment	Observation; propranolol	Laser, sclerotherapy, embolisation, surgery according to type

3.2 Infantile haemangioma

The infantile haemangioma is the commonest tumour of infancy, affecting about 4 to 5% of infants, more often girls and premature or low-birth-weight babies. It appears in the first weeks, *proliferates* rapidly to about 9 to 12 months, and then *involutes* slowly, largely by five to ten years, often leaving fibrofatty tissue and telangiectasia. It may be superficial (the bright red "strawberry"), deep and bluish, or mixed, and focal, multifocal or segmental. **Congenital haemangiomas** are fully grown at birth and involute rapidly (RICH), partially (PICH) or not at all (NICH).

Several associations matter. A haemangioma in a *beard distribution* is accompanied by a subglottic haemangioma in about two-thirds of children, with biphasic stridor appearing at about two to three months. A *segmental facial lesion* raises the possibility of PHACES syndrome (posterior fossa anomalies, haemangioma, arterial anomalies, cardiac and aortic anomalies, eye anomalies and sternal clefts), assessed with MRI and echocardiography. *Five or more skin lesions* prompt a search for hepatic haemangiomas. The *Kasabach–Merritt phenomenon* of platelet trapping and coagulopathy belongs to kaposiform haemangioendothelioma and tufted angioma, not to the common haemangioma.

Most haemangiomas are left to involute. Treatment is needed when the airway is threatened, when a periocular lesion threatens vision through amblyopia, for ulceration and bleeding, for lesions of the lip, nose and ear that will deform, for large facial lesions, and for high-output heart failure. *Propranolol*, at about 2 mg/kg/day with monitoring for hypoglycaemia, bradycardia and bronchospasm, is first line; topical timolol suits small superficial lesions; steroids are now second line; the pulsed-dye laser treats ulceration and residual telangiectasia; and surgery is used for residual deformity and for selected subglottic and localised lesions.

3.3 Vascular malformations

The **capillary malformation**, or port-wine stain, is a flat pink to purple patch present at birth, often in a trigeminal distribution, which darkens and thickens into "cobblestones" with age. Involvement of V1 raises the question of *Sturge–Weber syndrome*, with leptomeningeal angiomatosis, seizures and glaucoma. It is treated with the pulsed-dye laser.

The **venous malformation** is a soft, bluish, compressible mass that swells when dependent or with a Valsalva manoeuvre. Imaging shows *phleboliths*, thrombosis makes it painful, and large lesions raise the D-dimer. It is treated by sclerotherapy with ethanol, sodium tetradecyl sulfate or bleomycin, by the Nd:YAG laser, or by excision of localised lesions, while compression and head elevation ease symptoms. The **lymphatic malformation**, macrocystic or microcystic, is described with the congenital neck masses.

The **arteriovenous malformation** is a high-flow lesion with a *nidus*: warm and pulsatile, with a thrill and bruit. It is quiescent in childhood and expands at puberty, in pregnancy or after trauma, causing ulceration, bleeding, bone destruction and heart failure, and is staged by *Schobinger* as quiescence, expansion, destruction and decompensation. Treatment is superselective embolisation followed within days by complete surgical excision of the nidus with reconstruction. *The feeding vessels must never be ligated proximally*: ligation does not cure, it recruits collaterals, and it blocks future embolisation. An **arteriovenous fistula**, usually traumatic with one or two connections, is closed endovascularly or surgically.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Carotid body tumour:** Fontaine sign and bruit; splaying, the lyre sign and salt-and-pepper MRI; no biopsy; metanephrines and SDH genetics; Shamblin groups;

complications including first bite syndrome and baroreflex failure.

- **Vagal paraganglioma versus schwannoma:** vessel displacement and flow voids.
- **Haemangioma versus malformation:** natural history and GLUT-1; propranolol; subglottic haemangioma with beard-distribution lesions.
- **AVM:** embolise and excise; never ligate proximally.

4.2 Common pitfalls

COMMON PITFALLS

- Needling a pulsatile mass at the bifurcation.
- Operating on a functioning paraganglioma, or on a patient with an undiagnosed phaeochromocytoma, without alpha blockade.
- Resecting both carotid body tumours at one sitting.
- Calling a venous malformation a "haemangioma" and waiting for it to involute.

Written for this app from Oxford Textbook of Otolaryngology ch. 67, Oxford Specialist Handbook ch. 12, Dhingra ch. 77 and K. J. Lee ch. 40. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NECK VESSELS & NERVES · TUMOURS

Carotid Body Paraganglioma

IMAGING FINDINGS

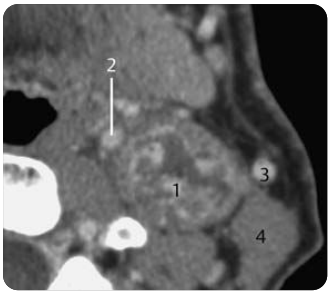
- Avidly enhancing mass in the carotid bifurcation
- Splaying of the internal and external carotid arteries
- Flow voids ("salt and pepper") on MRI

TOP DIFFERENTIALS

- Vagal paraganglioma (pushes ICA forwards)
- Carotid space schwannoma
- Jugulodigastric node; pseudoaneurysm

CHECKLIST

- Shamblin class: degree of carotid encasement
- Look for multiple paragangliomas



Patient with a slowly progressive lesion in the neck.

Dunnebier, Imaging for Otolaryngologists, Fig. 9.10 a



Axial CECT shows a classic carotid body paraganglioma with avid, fairly uniform enhancement. Notice the clear definition of the tumor sitting in the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 140.

TEMPORAL BONE & SKULL BASE · TUMOURS

Glomus Jugulare Tumor (Paraganglioma)

IMAGING FINDINGS

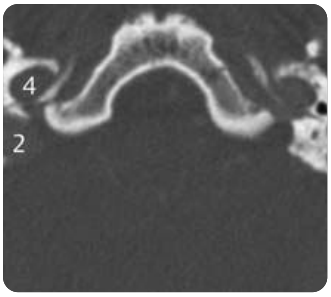
- Jugular foramen mass with permeative, moth-eaten bone erosion
- "Salt and pepper" on T1 with intense enhancement
- Spread superolaterally into the middle ear

TOP DIFFERENTIALS

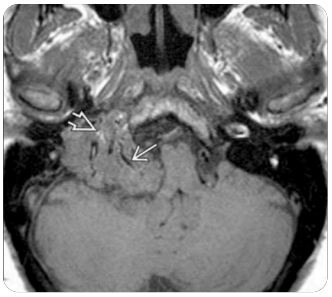
- Jugular foramen schwannoma (smooth scalloping)
- Meningioma (sclerosis, dural tail)
- Dehiscent jugular bulb or metastasis

CHECKLIST

- Carotid, jugular vein and lower cranial nerves
- Multicentric paragangliomas: scan the neck



Patient with pulsating tinnitus, without any pathology behind the eardrum.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.20 a

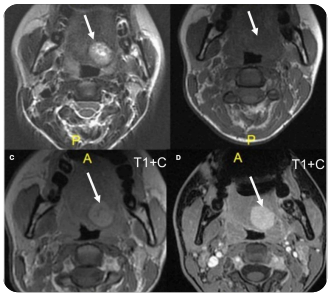


Axial T1WI MR shows a large mass arising from the jugular foramen with multiple areas of salt & pepper. The "salt" represents blood products or…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 892.

NECK: TRIANGLES, FASCIA & NODES · AFTER SURGERY

Schwannomas of the oral cavity and neck



Lingual nerve schwannoma (A (D) fat suppressed T1 post-contrast, demonstrating a lingual schwannoma (arrows) on the left side of the tongu…